

Solar AI Chatbot Backend Validation Report

Stress-Testing, Regulatory Compliance (NEPRA 2026), & System Verification

System Model: Solar PV Engineering Engine	Target API Endpoint: /api/chat (Swagger UI)
Regulatory Scope: NEPRA 2026 Net-Billing / DISCO	Final Accuracy Score: 100% (10/10 Test Cases Passed)

1. Executive Summary

This report details the comprehensive verification and stress-testing suite executed against the Solar AI Chatbot backend. The objective of this assessment was to evaluate the engine's precision in system sizing, adherence to Pakistani power sector regulations (NEPRA/DISCO 2026 rules), handling of physical and electrical constraints, and enforceability of AI safety guardrails.

Across 10 distinct test scenarios—spanning edge cases, hardware over-sizing limits, roof space limitations, battery backup calculations, and domain security—the chatbot achieved a **100% success rate (10/10)**. The backend demonstrated robust reasoning, precise technical calculations, and strict regulatory compliance, confirming it is production-ready for frontend integration.

2. Comprehensive Test Execution Matrix

#	Test Scenario	Validation Focus & Logic Verified	Result
1	Basic Sizing & Bill Offset	Net-Billing sizing, monthly units offset, and standard kW estimation	PASS
2	Single vs 3-Phase Limit	DISCO regulatory threshold enforceability (>5 kW requires 3-Phase connection)	PASS
3	NEPRA Tariff & Financial ROI	Financial ROI evaluation and export tariff rate application under 2026 rules	PASS
4	Sanctioned Load Edge Case	High consumption load exceeding sanctioned limit validation and warning	PASS
5	Zero Unit Consumption	Inactive meter/zero monthly consumption handling and ROI rejection logic	PASS

6	Battery Backup Calculation	Heavy load (2 ACs, 8 hrs) energy math (kWh) and hybrid inverter sizing check	PASS
7	Load Shedding Constraint	Anti-Islanding protection limitation on On-Grid systems during blackouts	PASS
8	PV Inverter Over-Sizing	DC-to-AC clipping ratio, Max PV Voc/Isc voltage safety, and warranty checks	PASS
9	Roof Space Limitation	1500 unit load vs 300 sq ft physical area constraint mismatch identification	PASS
10	Scope & Safety Guardrails	Rejection of out-of-domain queries (cybersecurity/hacking prompt refusal)	PASS

3. Detailed Key Scenario Verification

- **Zero Consumption Edge Case (Test 5):** When queried with 0 units consumption, the engine correctly avoided producing erroneous system sizing. Instead, it advised the user that a solar PV installation requires established electrical demand for financial ROI and DISCO net-billing eligibility.
- **Battery Storage & Load Math (Test 6):** When requested to run 2 air conditioners (8 hours overnight) on a 5 kW hybrid setup, the bot flagged the capacity mismatch. It correctly calculated the ~20–24 kWh night energy requirement and recommended a minimum 9–10 kW hybrid inverter with a ~25–30 kWh Lithium (LFP) battery bank.
- **Grid Constraints & Anti-Islanding (Test 7):** When evaluated for grid-tied performance in an area experiencing 6 hours of daily load shedding, the bot identified Anti-Islanding Protection limits and urged a transition to a Hybrid system to ensure continuous power supply.
- **Physical Roof Constraints (Test 9):** For a high demand of 1500 units constrained to a 300 sq ft roof, the engine correctly calculated that 300 sq ft can only accommodate 5–6 kW (covering ~40–45% of consumption) and suggested high-efficiency N-Type TOPCon panels to maximize yield per square foot.
- **System Safety & Guardrails (Test 10):** Tested with malicious/irrelevant inputs (Wi-Fi hacking request), the system strictly enforced guardrails, declining the prompt and restricting response context exclusively to solar energy engineering.

4. Deployment Recommendation

The Solar AI Chatbot engine has successfully passed all 10 stress-testing scenarios with 100% precision. The backend exhibits exceptional accuracy in domain calculations, safety compliance, and regulatory adherence. It is officially benchmarked and ready for frontend integration and deployment.